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Designing a Magnetic Gear for an Electric Aircraft Drivetrain

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Abstract— This paper presents the analysis and mechanical design of a 15.9kW magnetic gear (MG) with a 12.4:1 gear ratio and a computed 53 N·m/kg mass torque density. When the MG is connected in series with a high-speed motor it forms an aircraft MG drivetrain. The target specifications are based on the quad-electric aircraft propulsion system needed for hover-load conditions for NASA's electric vertical take-off and landing vehicle.

Keywords—Aircraft drivetrain, high torque density, high gear ratio, magnetic gear

I. INTRODUCTION

In order to develop the next generation of fixed wing and vertical-lift electric aircraft, the aircraft efficiency, emissions, reliability, and noise needs to be improved [1]. Consequently, the aircrafts drivetrain needs to operate at a very high mass torque density whilst also being exceptionally reliable, efficient, robust, and environmentally benign.

In order to make an aircraft drivetrain more compact mechanical gears are often used [2]. However, the use of mechanical gears within an aircraft introduces a number of system level consequences such as the need for a lubrication system to reduce tooth friction and remove heat [3], [4]. A back-up lubrication system is then also required to meet certification requirements in the advent that there is a loss of lubrication performance. And the gearing systems introduce significant cabin noise due to the gear-tooth vibrations [5]. Furthermore, to mitigate tooth wear and failure, the aircraft needs to be grounded periodically for inspection and maintenance [3], [4].

One of the main alternatives to using mechanical gearing is the use of a direct-drive permanent magnet (PM) motor [2], [6], [7]. The use of direct-drive PM motor removes the reliability concerns with respect to the mechanical gearbox. However, the torque density of a direct-drive PM motor is thermally limited (by current) and therefore a PM motor does not normally achieve torque densities greater than 50 N·m/L [8]–[11] and 10.6 N·m/kg [12]. For this reason, direct-drive PM generators become very large when scaled-up in size.

The use of a magnetic gear (MG) within an aircraft drivetrain could potentially offer improvement capabilities by being able to offer the reliability of direct-drive generators whilst being almost as compact as a mechanically geared drivetrain [1], [13]. Magnetic gears (MGs). A number of papers

have shown that MGs can be designed with efficiencies close to 99% [14]. An illustration of a coaxial MG with a Halbach rotor typology is shown in Fig. 1. By selecting the number of modulator slots, n_2 , to be equal to $n_2 = p_1 + p_3$, where p_1 = inner rotor pole-pairs, and p_3 = outer rotor pole-pairs then the different inner and outer rotor field space harmonics are coupled via the slotting of the central ferromagnetic rotor. If the outer rotor is held fixed, the gear ratio, G_r , between the inner high-speed rotor and slotted low-speed cage rotor will be [15]

$$G_r = \frac{n_2}{p_1} \quad (1)$$

The majority of MG designs constructed to-date have been focused on maximizing volumetric torque density [16]–[22]. Whereas, for aerospace applications, mass is the dominant design factor. The MG model mass torque density can be computed using

$$T_m = \frac{T_2}{m_s + m_m} \quad (2)$$

where T_2 is the low-speed rotor torque, m_s = structural mass and m_a = active region mass (magnets and laminations). This paper presents the design of a MG which has a total mass torque density

TABLE I TARGET SPECIFICATIONS

Metric	Value	Units
Gear ratio	≥ 12.1	-
Rated power	15.9	kW
Rated torque, T_2	229	N·m
Peak efficiency	> 98	%
Mass torque density	≥ 50	N·m/kg

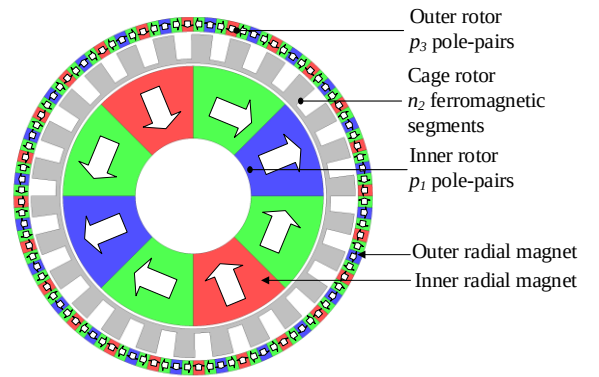


Fig. 1. Coaxial MG with Halbach array PMs and $p_1 = 2$ pole-pairs, $n_2 = 26$ steel poles and $p_3 = 24$ pole-pairs.

objective above 50 N·m/kg. All the target performance metrics are listed in Table I. Using (2), the require total MG mass $m_t = m_s + m_m$ must be $m_t \leq 4.32$ kg. The target specifications for the MG drivetrain are based on the quad-electric aircraft propulsion system needed for hover-load conditions for NASA's electric vertical take-off and landing reference vehicle [23].

II. POLE-COMBINATION ANALYSIS

A review of the pole-pair combinations that can yield a gear ratio between $G_r = 12$ and $G_r = 13$ are shown in Fig. 2. The following criteria was used to select the pole combination to use:

- The cage rotor slot number, n_2 , should be an even number which allows an evenly distributed radial force on the modulator rotor that is able to minimize radial deflection.
- A non-integer gear ratio should be selected as this leads to a lower torque ripple and noise.
- To mitigate harmonic interaction the inner or outer rotor base pole-pair number must be selected to be a prime number.

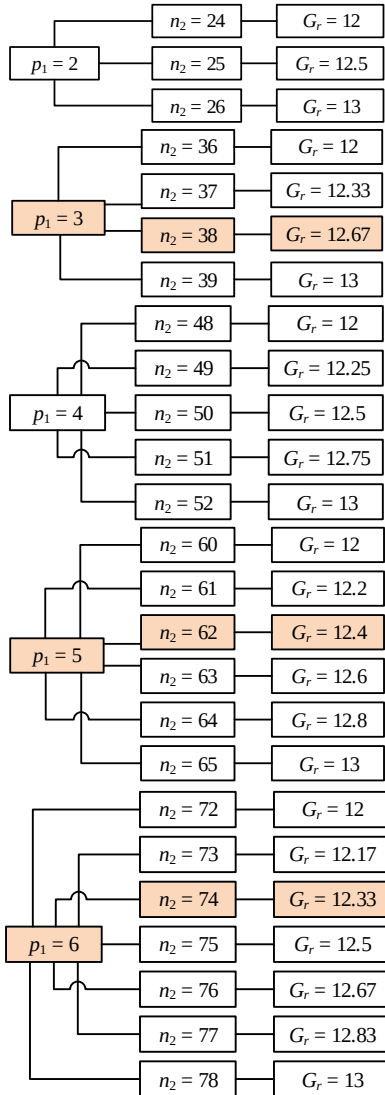


Fig. 2. Considered pole-slot combinations. The selected design cases: *Design I*: $(p_1, n_2, p_3) = (5, 62, 57)$, *Design II*: $(p_1, n_2, p_3) = (3, 38, 35)$, *Design III*: $(p_1, n_2, p_3) = 2 \times (3, 37, 34)$.

Based on the above criteria the three pole-combination designs highlighted in Fig. 2 were selected for further analysis. Note that *Design III* has a base pole combination of $(p_1, n_2, p_3) = (3, 37, 34)$ this combination satisfies the prime number criteria. But this combination has an odd number of cage slots, therefore, a factor of two was applied to the base combination, yielding a pole combination of $(p_1, n_2, p_3) = (6, 74, 68)$. As the harmonics are odd the same harmonic interaction occurs.

III. ELECTROMAGNETIC FINITE ELEMENT ANALYSIS

The MG design and fixed geometric parameters that are being utilized in the sweep analysis are shown in Fig. 3. and Table II. The inner and outer rotors have a 4-segment Halbach rotor typology and the modulator slots are mechanically supported using the inner rotor bridges and rectangular Garolite rods (not shown). As the radial force components on the modulator slots are primarily radially outward (positive), as illustrated in Fig. 4, the rectangular Garolite cage rods are supported in place using epoxy glue. The sweep parameters used in the initial sweep analysis for each pole-combination selection are shown in Table III. The selected outer radii were chosen to allow the outer rotor magnets to have a minimum width of $w_3 = 3$ mm and the stack length was selected to ensure sufficient torque when the edge-effect reduction is low. The cage rotor length was kept fixed at a minimum value of $l_2 = 5$ mm, this radial thickness was initially determined to be the

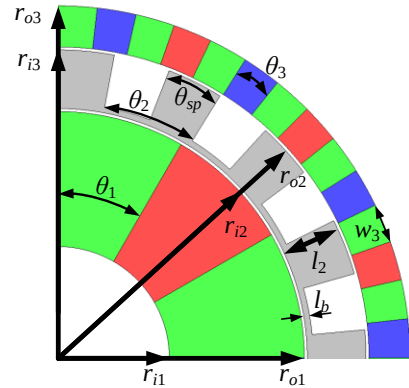


Fig. 3. Geometric parameter definitions.

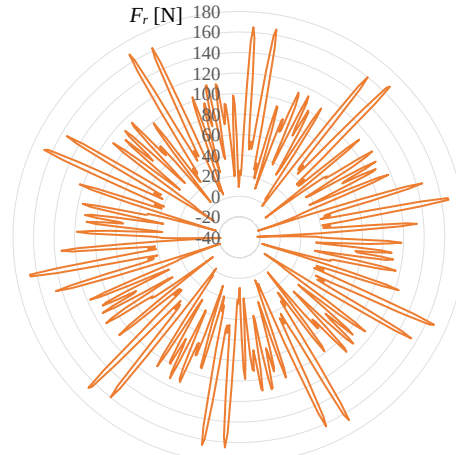


Fig. 4. Radial force, F_r , on one modulator segment as it rotated around by 360° when both the inner and outer rotors are held fixed.

TABLE II FIXED GEOMETRIC PARAMETERS

Description	Value	Unit
Inner rotor	Angular span, θ_1	$\pi/(4p_1)$ radians
Cage rotor	Angular span, θ_2	$\pi/(n_2)$ radians
	Angular span, θ_{sp}	$\pi/(2p_3)$ radians
Outer rotor	Angular span, θ_3	$\pi/(4p_3)$ radians
Outer rotor airgap, g_o	0.5	mm
Inner rotor airgap, g_i	0.86	mm
Modulator bridge thickness, l_b	0.75	mm
Magnet grade	N50H	-
Cage lamination material	Sura M270 50A	-
Cage lamination filling factor	95%	-

TABLE III. INITIAL PARAMETER SWEEP VALUES

Design #	I	II	III
Pole combination (p_1, n_2, p_3)	3, 38, 35	5, 62, 57	6, 74, 68
Gear ratio, G_r	12.67	12.4	12.33
Inner rotor inner radius, r_{i1} [mm]	[50, 51, ..., 71]	[75, 76, ..., 96]	[105, 106, ..., 126]
Inner rotor outer radius, r_{o1} [mm]	[60, 61, ..., 73]	[95, 96, ..., 105], 99.5	[115, 116, ..., 136]
Cage radial length, l_2 [mm]	5	5	5
Outer rotor outer radius, r_{o3} [mm]	80	110	135
Axial stack length, d [mm]	60	30	30

TABLE IV 2-D SWEEP PARAMETERS

Pole combination (p_1, n_2, p_3)	5, 62, 57
Outer rotor outer diameter, r_{o3}	[100, 111] mm
Inner rotor inner radius, r_{i1}	[75, 77, ..., 95] mm
Inner rotor outer radius, r_{o1}	[95, 96, ..., 105], 99.5 mm
Cage radial length, l_2	[3, 4, ..., 7] mm
Axial stack length, d	30 mm

minimum possible that could be easily fabricated. A bridge thickness of $l_b = 0.75\text{mm}$ was used. This was the minimum bridge thickness that could be provided by the manufacturer. The inner airgap was made larger than the outer airgap so as to provide space for a 0.36 mm thick Kevlar band around the inner rotor magnets.

When relying on 2-D modelling, and during the initial geometric sweep analysis, an assumed axial edge-effect factor $k_e = 0.7$ was used. This factor approximates the torque reduction when converting the model from 3-D to 2-D. The axial edge-effects are caused by the flux leakage. Preliminary simulations were conducted to ensure that a 30% reduction was appropriate for all three combinations.

Using the edge effect factor the required 2-D torque is calculated to be

$$T_2^{2D} = \frac{T_2}{k_e} = 327 \text{ N} \cdot \text{m} \quad (3)$$

The 2-D parameter sweep analysis results are shown in Fig. 5, the plot shows that *Design II* and *Design III* have the lowest active region mass, m_a .

However, due to the large outer diameter for *Design III*, the mass required for the mechanical structure will be greater, hence, *Design II*, with parameters $(r_{i1}, r_{o1}) = (91, 99.5)$ was

selected, giving an active material mass of $m_a = 2.3 \text{ kg}$. A complete parameter sweep for *Design II* was then conducted using the 2-D and 3-D iterative FEA sweep procedure outlined in Fig. 6. The sweep parameters values used in the iterative analysis are shown in Table IV.

The flow chart, shown in Fig. 6, was repeated for the following stack lengths $d = [20, 22.5, \dots, 30] \text{ mm}$. Designs with appropriate torque and the lowest mass associated with each stack length are listed in Table V. *Design D* yielded the smallest active mass, m_a , however, *Design C* was selected after considering manufacturing limitations. The performance map indicating the scaled 2-D torque and active mass for the final iteration when the stack length $d = 25 \text{ mm}$ was used is shown in Fig. 7. Also shown is the location of *Design C*. The selected *Design C* was then amended by reducing r_{i1} by 1mm so that the MG design achieved $T_2^{3D} = 240.2 \text{ N} \cdot \text{m}$ at an operational temperature of 40°C and with an active mass of 2.5 kg . The finalized magnetic design is illustrated in Fig. 8 and the geometric parameters and 3-D FEA calculated performance values are listed in Table V. The calculated active volumetric and mass torque density are $248 \text{ N} \cdot \text{m/L}$ and $95 \text{ N} \cdot \text{m/kg}$ respectively.

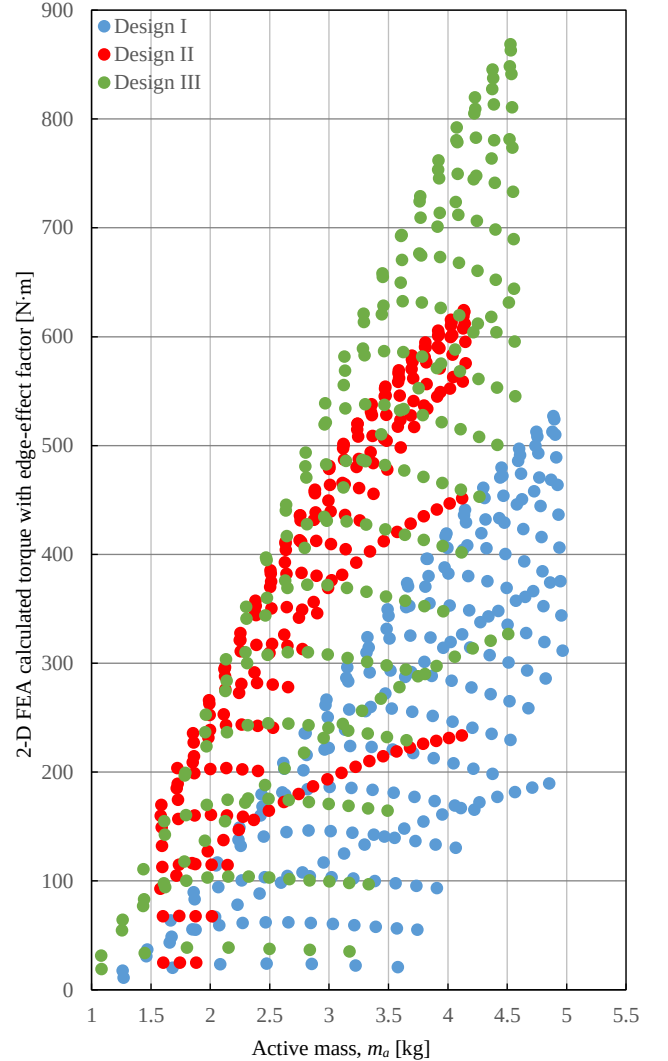


Fig. 5. Combined performance analysis results for the three MG designs.

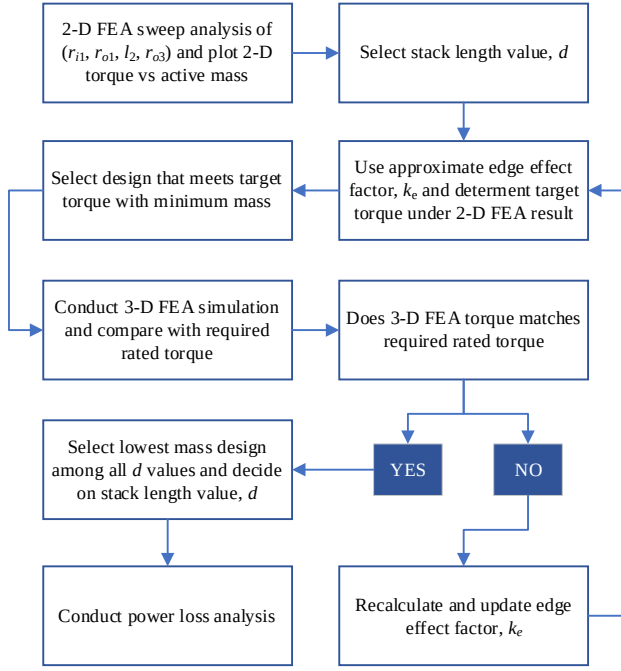


Fig. 6. Iterative finite element analysis electromagnetic design procedure.

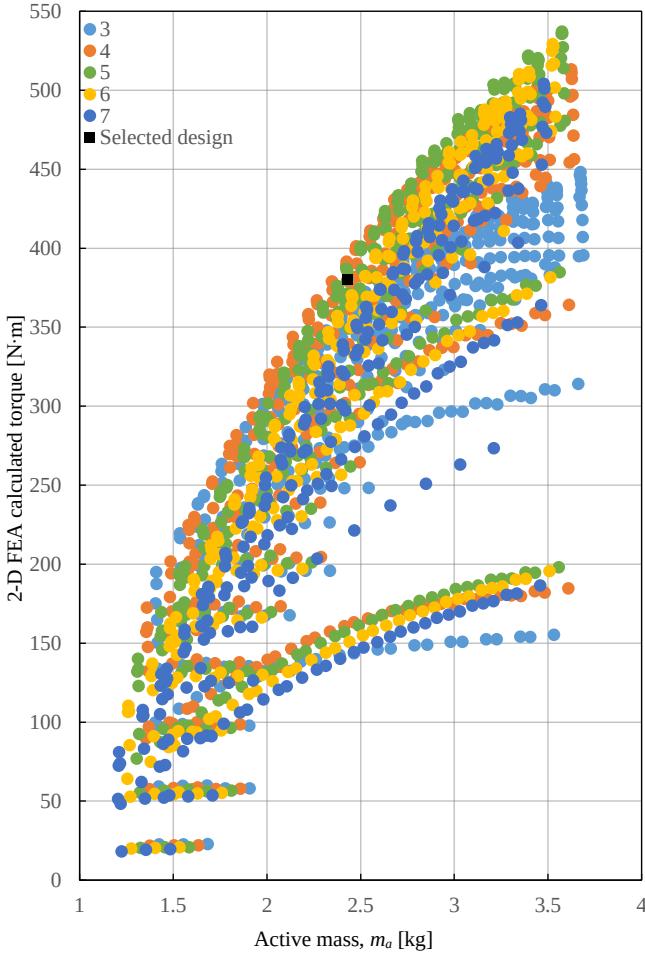


Fig. 7. Geometric sweep plot for the $(p_1, n_2, p_3) = (5, 62, 57)$ pole-slot combination. The plot shows the active mass for each design case and the 2-D FEA computed torque. The legend shows different l_2 .

IV. EXPERIMENTAL PROTOTYPE

A cut-away view of the structural design for the MG is shown in Fig. 9. The fabricated inner rotor with Halbach array magnets is shown in Fig. 10. Bonding glue and Kevlar threads were used to secure the magnets. An inner rotor field comparison between the measured and computed values is shown in Fig. 11. The harmonic analysis indicates that the radial magnetic flux density is 1.4 % lower than the 3-D FEA calculated value.

TABLE V. FLOW CHART ITERATION CONCLUSION FOR DIFFERENT STACK LENGTH AT 20°C

Parameters and performance variables	Designs					Units
	A	B	C	D	Final	
Stack length, d	20	22.5	25	27.5	25	mm
Axial edge-effect factor, k_e	0.55	0.585	0.631	0.66	0.62	-
Inner rotor inner radius, r_{i1}	75	81	87	87	86	mm
Inner rotor outer radius, r_{o1}	99.5	99.5	99.5	98	99.5	mm
Cage radial length, l_2	5	5	5	5	5	mm
Outer rotor outer radius, r_{o3}	111	110	111	109	111	mm
2-D Torque, T_2^{2D}	429.6	414.2	379.8	364.7	405.0	N·m
3-D Torque, T_2^{3D}	236.2	242.4	240.1	241.8	251.0	N·m
Active mass, m_a	2.86	2.6	2.43	2.39	2.53	kg

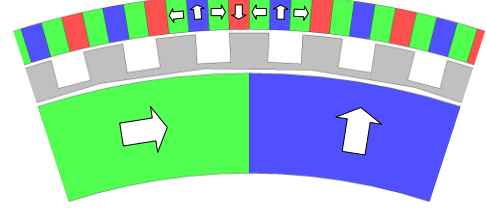


Fig. 8. Sectional drawing of final magnetic design.

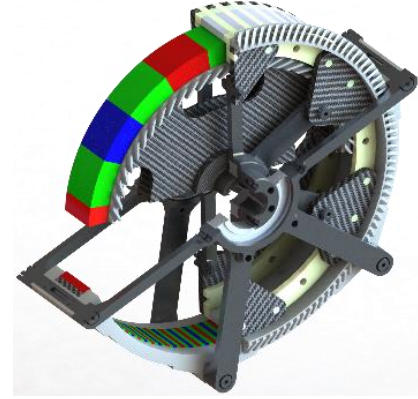


Fig. 9. Sectional view of the mechanical assembly for the MG with carbon fiber rotor yoke.

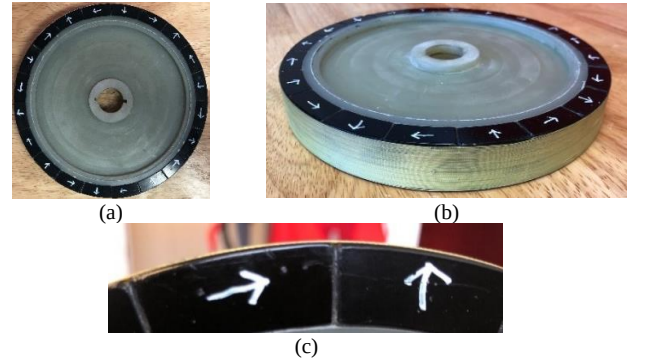


Fig. 10. (a) Top view of the fully fabricated inner rotor and (b) side view showing added Kevlar band and (c) edge view.

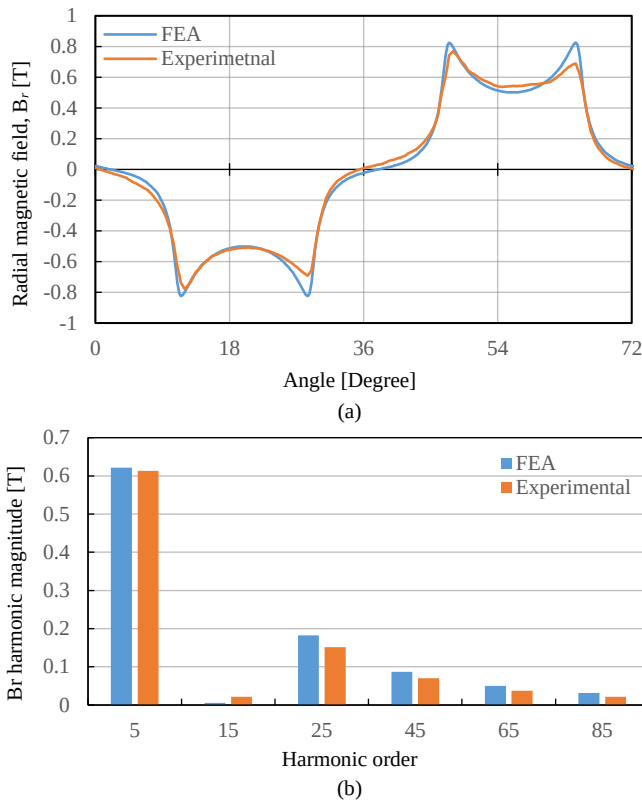


Fig. 11. Inner rotor radial magnetic (a) field and (b) field harmonic component comparison between FEA simulation result and measurement at 0.95 mm above inner rotor surface.

According to the target specifications, inner rotor needs to operate at a peak angular speed of 8196 r/min. Based on the final design geometry, a combined centrifugal and magnetic force of 6.5 kN is present at the peak speed and this results in a peak radial stress of 9.57 MPa. Loctite E-40HT was selected and tested on the setup shown in Fig. 12, it was measured to provide a peak stress of 23.4 MPa. This demonstrated a safety factor of 2.4. To further enhance the mechanical support a Kevlar thread shown in Fig. 10 was used.

Fig. 13 shows the assembled modulator rotor with the rectangular Garolite G11 rods fully inserted. All support rods were glued in place with Loctite E-40HT as well. Note that to reduce fabrication costs for the proof-of-principle design the rotor support parts were fabricated using Garolite rather than carbon-fiber. Due to the Corona virus, the MG construction and testing was not completed at the time of this paper submission.

V. CONCLUSION

This paper presented a parameter sweep analysis and mechanical design details for an electric aircraft MG that used a Halbach rotor typology. The mass torque-density, rather than volumetric torque density was a key design criteria. The designed MG achieved a calculated peak torque of 240 Nm with a 12.4:1 gear ratio. If carbon fiber support parts were used a mass torque density of $T_m = 53 \text{ N}\cdot\text{m}/\text{kg}$ was calculated to be achievable.

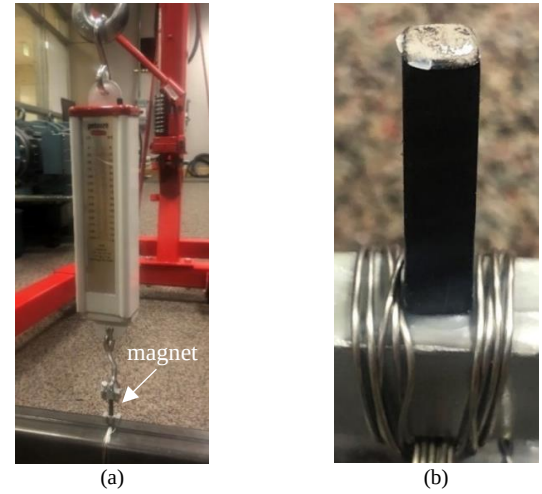


Fig. 12.(a) Loctite E-40HT glue testing setup and (b) damaged magnet with epoxy coating peeled off.

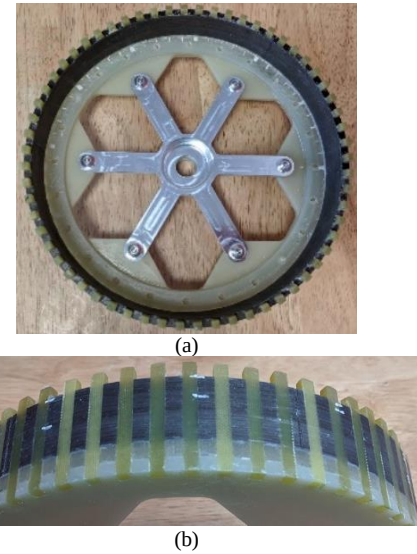


Fig. 13. (a) Top view showing fully assembled modulator rotor with rectangular Garolite G11 rods fully inserted, (b) zoomed in view showing rod supports connected to endplate.

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REFERENCES

- [1] V. Asnan, J. Scheidler, and T. Talerico, "Magnetic gearing research at NASA," presented at the AHS International 74th Annual Forum & Technology Display, Phoenix, Arizona, USA, May 2018.
- [2] A. D. Anderson *et al.*, "System Weight Comparison of Electric Machine Topologies for Electric Aircraft Propulsion," in *2018 AIAA/IEEE Electric Aircraft Technologies Symposium (EATS)*, Jul. 2018, pp. 1–16.
- [3] "Rotorcraft Drivetrain Life Safety and Reliability," North Atlantic Treaty Organization Advisory Group For Aerospace Research And Development, 775, Jun. 1990.
- [4] E. V. Zaretsky, "Bearing and Gear Steels for Aerospace Applications," NASA, Mar. 1990.

- [5] J. W. Titus, "Analysis of Friction Torque in Simple and Preloaded Spur Gear Trains," *Nav. Res. Lab Wash. DC*, 1965.
- [6] X. Zhang, C. L. Bowman, T. C. O'Connell, and K. S. Haran, "Large electric machines for aircraft electric propulsion," *IET Electr. Power Appl.*, vol. 12, no. 6, pp. 767–779, 2018, doi: 10.1049/iet-epa.2017.0639.
- [7] A. Borisavljevic, "Limits, Modeling, and Design of High-Speed Permanent Magnet Machines," *Springer*, vol. 2013, doi: ISBN 978-3-642-33457-3.
- [8] Shi Jinhao, Li Yongbin, and Zheng Wenpeng, "Survey on the development of transverse flux machines," in *Sixth International Conference on Electrical Machines and Systems, 2003. ICEMS 2003.*, Nov. 2003, vol. 1, pp. 137–140 vol.1.
- [9] M. Dubois, "Optimized Permanent Magnet Generator Topologies for Direct-Drive Wind Turbines," Jan. 2004.
- [10] P. Kasinathan, A. Grauers, and E. S. Hamdi, "Force density limits in low-speed permanent-magnet machines due to saturation," *IEEE Trans. Energy Convers.*, vol. 20, no. 1, pp. 37–44, Mar. 2005, doi: 10.1109/TEC.2004.837302.
- [11] D. G. Dorrell, S. S. Ngu, and C. Cossar, "Comparison of High Pole Number Ultra-Low Speed Generator Designs Using Slotted and Air-Gap Windings," *IEEE Trans. Magn.*, vol. 48, no. 11, pp. 3120–3123, Nov. 2012, doi: 10.1109/TMAG.2012.2201704.
- [12] "Electric propulsion components with high power densities for aviation," presented at the Transformative Vertical Flight Workshop, Aug. 2015.
- [13] J. J. Scheidler, V. M. Asnani, and T. F. Tallerico, "NASA's Magnetic Gearing Research for Electrified Aircraft Propulsion," in *2018 AIAA/IEEE Electric Aircraft Technologies Symposium (EATS)*, Jul. 2018, pp. 1–12.
- [14] T. Tallerico, J. Scheidler, and Z. Cameron, *Electromagnetic Mass and Efficiency of Magnetic Gears for Electrified Aircraft*. 2019.
- [15] K. K. Uppalapati, W. B. Bomela, J. Z. Bird, M. D. Calvin, and J. D. Wright, "Experimental Evaluation of Low-Speed Flux-Focusing Magnetic Gearboxes," *IEEE Trans. Ind. Appl.*, vol. 50, no. 6, pp. 3637–3643, Dec. 2014, doi: 10.1109/TIA.2014.2312551.
- [16] K. K. Uppalapati, M. D. Calvin, J. D. Wright, J. Pitchard, W. B. Williams, and J. Z. Bird, "A Magnetic Gearbox With an Active Region Torque Density of 239 N·m/L," *IEEE Trans. Ind. Appl.*, vol. 54, no. 2, pp. 1331–1338, Apr. 2018, doi: 10.1109/TIA.2017.2779418.
- [17] M. Desvaux *et al.*, "Magnetic losses and thermal analysis in a magnetic gear for wind turbine," in *2018 Thirteenth International Conference on Ecological Vehicles and Renewable Energies (EVER)*, Apr. 2018, pp. 1–7, doi: 10.1109/EVER.2018.8362382.
- [18] M. Johnson, M. C. Gardner, H. A. Toliyat, S. Englebreton, W. Ouyang, and C. Tschida, "Design, Construction, and Analysis of a Large-Scale Inner Stator Radial Flux Magnetically Geared Generator for Wave Energy Conversion," *IEEE Trans. Ind. Appl.*, vol. 54, no. 4, pp. 3305–3314, Aug. 2018, doi: 10.1109/TIA.2018.2828383.
- [19] B. McGilton, R. Crozier, A. McDonald, and M. Mueller, "Review of magnetic gear technologies and their applications in marine energy," *IET Renew. Power Gener.*, vol. 12, no. 2, pp. 174–181, Summer 2018, doi: 10.1049/iet-rpg.2017.0210.
- [20] T. V. Frandsen, P. O. Rasmussen, and K. K. Jensen, "Improved motor integrated permanent magnet gear for traction applications," in *2012 IEEE Energy Conversion Congress and Exposition (ECCE)*, Sep. 2012, pp. 3332–3339, doi: 10.1109/ECCE.2012.6342334.
- [21] L. Jian, K. T. Chau, and J. Z. Jiang, "An integrated magnetic-g geared permanent-magnet in-wheel motor drive for electric vehicles," in *2008 IEEE Vehicle Power and Propulsion Conference*, Sep. 2008, pp. 1–6, doi: 10.1109/VPFC.2008.4677512.
- [22] H. Y. Wong, J. Z. Bird, D. Barnett, and W. Williams, "A High Torque Density Halbach Rotor Coaxial Magnetic Gear," in *2019 IEEE International Electric Machines & Drives Conference (IEMDC)*, May 2019, pp. 233–239, doi: 10.1109/IEMDC.2019.8785188.
- [23] W. Johnson, C. Silvia, and E. Solis, "Concept Vehicles for VTOL Air Taxi Operations." American Helicopter Society Technical Conference.